Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

LESSON PLAN

For

Autonomous AI Agents with OpenClaw

TGS Ref No: TGS-2026064859

Conducted by

Tertiary Infotech Academy Pte Ltd

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Version 6.0

DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
4.0	23 August 2026	Rebuilt the 120-slide deck around a detailed OpenClaw architecture spine, restored the substantive original concepts, expanded each lab into concept/configuration/ diagnosis slides, and added ten worked use cases while preserving the approved schedule.	Tertiary Infotech Academy Pte Ltd
6.0	23 August 2026	Replaced conceptual architecture summaries with implementable protocol, configuration, persistence, policy, observability and recovery contracts while preserving the approved schedule, ten lab units and ten detailed use-case implementations.	Tertiary Infotech Academy Pte Ltd

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Course Title	Autonomous AI Agents with OpenClaw
Course Code	TGS-2026064859
TGS Ref No	TGS-2026064859
Duration	1 Day (8 hours, including 1-hour assessment)
Schedule	09:00 AM – 06:00 PM | Tea break 10:20 | Lunch 12:45–13:45
Delivery Mode	Instructor-led, hands-on labs
Environment	Option A: Hostinger VPS (Ubuntu 22.04) OR Option B: Docker Desktop
Institution	Tertiary Infotech Academy Pte Ltd
Instructor	Mohan Pothula

Course Overview

This 1-day hands-on WSQ course teaches participants to deploy and operate OpenClaw, an open-source AI agent framework, on a Hostinger VPS or local Docker Desktop. The 119-slide implementation-level deck explains the Gateway, agent loop, sessions, context engine, compaction, memory, tools, policy layers, channels, automation, multi-agent routing and responsible autonomy, then applies ten worked use cases from trigger through evidence. The morning combines theory and setup labs 1–3 before lunch. The afternoon moves through practice labs 4–9 covering tools, skills, cron jobs, security hardening, the web dashboard, and cost optimisation.

Learning Outcomes

By the end of this course, participants will be able to:

- LO1: Deploy OpenClaw on a Hostinger VPS or Docker Desktop local machine
- LO2: Connect LLM providers — Groq (free), OpenAI, Ollama, or Default config
- LO3: Set up Telegram (BotFather) and WhatsApp (QR pairing) messaging channels
- LO4: Integrate AgentMail, Agent Browser, and Firecrawl tools
- LO5: Install and invoke skills from the ClawHub marketplace
- LO6: Schedule tasks with cron jobs and monitor uptime with heartbeat
- LO7: Apply 10-step security hardening for OpenClaw
- LO8: Access the OpenClaw web dashboard locally and via SSH tunnel
- LO9: Optimise LLM costs using model selection and context compaction
- LO10: Implement blockchain invoice verification using Docker and OpenClaw

Prerequisites

- Laptop: Windows 10/11 (WSL2), macOS 12+, or Ubuntu 22.04+
- Docker Desktop installed (required for Option B and Lab 10)
- Admin / sudo rights on the machine
- Telegram account + phone with WhatsApp installed
- Free Groq API key at https://console.groq.com (no credit card required)

- Free AgentMail account at https://agentmail.to
- Free Firecrawl account at https://www.firecrawl.dev

1-Day Schedule

Morning Session · 09:00 AM – 12:45 PM

Time	Topic / Activity	Duration
09:00 – 09:20	Welcome, Digital Attendance & Course Overview	20 min
09:20 – 09:50	**Topic 1 — Overview of OpenClaw: architecture, use cases & diagrams**	30 min
09:50 – 10:20	**Topic 3 — Tools, Skills & Channels Overview**	30 min
10:20 – 10:35	***Tea Break***	15 min
10:35 – 10:55	**Topic 4 — Security, Cost Saving & Best Practices**	20 min
10:55 – 11:15	**Topic 2 — OpenClaw Applications + Blockchain Demo**	20 min
11:15 – 11:45	Lab 1: OpenClaw Hosting (Hostinger VPS or Docker Desktop)	30 min
11:45 – 12:15	Lab 2: OpenClaw Model (Groq free / OpenAI / Ollama)	30 min
12:15 – 12:45	Lab 3: OpenClaw Channel (Telegram + WhatsApp)	30 min

Time	Topic / Activity	Duration
12:45 – 13:45	***Lunch Break***	60 min

Afternoon Session · 01:45 PM – 06:00 PM

Time	Topic / Activity	Duration
13:45 – 14:15	Lab 4: OpenClaw Tools (AgentMail / Agent Browser / Firecrawl)	30 min
14:15 – 14:35	Lab 5: OpenClaw Skills (ClawHub marketplace)	20 min
14:35 – 15:05	Lab 6: Cron Jobs and Heartbeat	30 min
15:05 – 15:35	Lab 7: OpenClaw Security (10-step hardening guide)	30 min
15:35 – 15:55	Lab 8: OpenClaw Dashboard (SSH tunnel for VPS users)	20 min
15:55 – 16:15	Lab 9: OpenClaw Cost Saving (model selection + compaction)	20 min
16:15 – 16:45	Lab 10: Blockchain Invoice Verification	30 min
16:45 – 17:00	Assessment Briefing and TRAQOM	15 min
17:00 – 18:00	**Written Assessment (SAQ) + Case Study**	60 min

Labs Covered

All 10 hands-on labs run on the student's chosen environment (Hostinger VPS or Docker Desktop):

Lab	Title	Duration
Lab 1	OpenClaw Hosting	30 min
Lab 2	OpenClaw Model	30 min
Lab 3	OpenClaw Channel	30 min
Lab 4	OpenClaw Tools	30 min
Lab 5	OpenClaw Skills	20 min
Lab 6	Cron Jobs and Heartbeat	30 min
Lab 7	OpenClaw Security	30 min
Lab 8	OpenClaw Dashboard	20 min
Lab 9	OpenClaw Cost Saving	20 min
Lab 10	Blockchain Invoice Verification	30 min

Assessment

Participants are assessed through hands-on lab work throughout the day and a final Written Assessment (Short Answer) + Case Study on the LMS (open book), completed within 1 hour. A minimum of 75% attendance is required for SSG funding eligibility. TRAQOM survey is mandatory.

Practical demonstration — students must show:

- openclaw gateway status → Gateway: running
- Agent responds to a message via Telegram and/or WhatsApp
- Firecrawl tool returns a webpage summary
- Cron job fires on schedule and sends result to channel
- Dashboard accessible at http://localhost:18789
- Blockchain: invoice INV-2026-001 registers and verifies via Docker on port 5000